CS & IT ENGINEERING

Digital Logic

Sequential Circuits

Lecture No. 04

By- Aditya sir

Recap of Previous Lecture

Topic

Latches

Flipflops

↳ 1) S-R FF (6 steps)

Topics to be Covered

Topic

1) J-K FF
2) D FF

About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.

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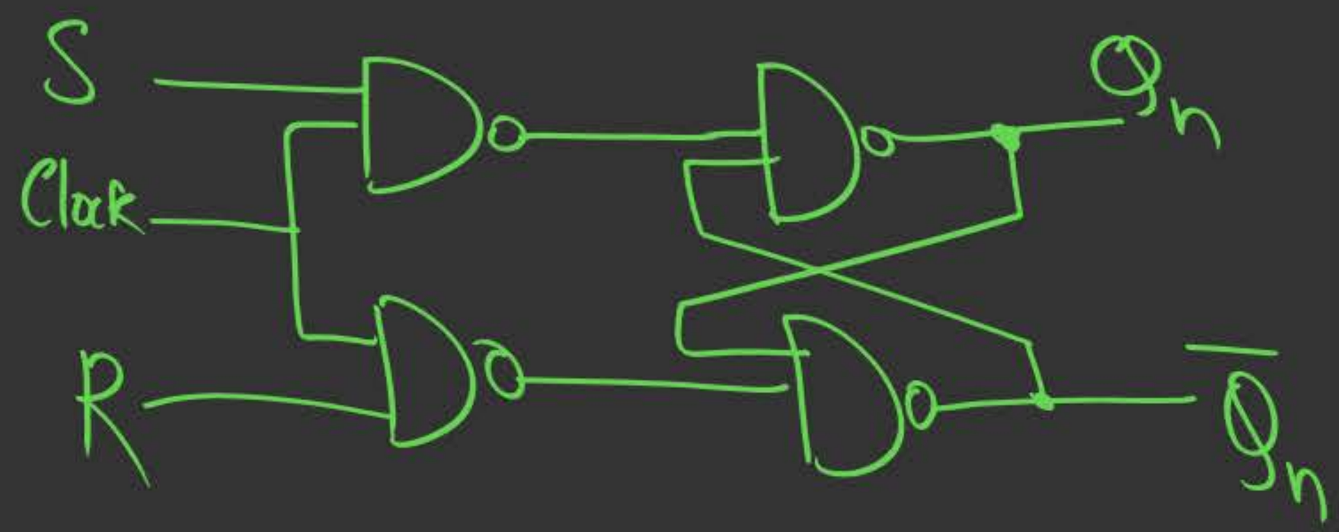
Recap of S-R FF

(for details watch prev lecture)

1) <u>Circuit Diagram</u>



2) <u>Truth table</u>

S	R	Q_{n+1}	$\overline{Q}_{n+1}$	
0	0	Q_n	$\overline{Q}_n$	→ Hold
0	1	0	1	→ Reset
1	0	1	0	→ Set
1	1	X	X	→ dont care

3) Characteristic table: $(S, R, Q_n \longrightarrow Q_{n+1})$

S	R	Q_n	Q_{n+1}	
0	0	0	0	0
0	0	1	1	1
0	1	0	0	2
0	1	1	0	3
1	0	0	1	4
1	0	1	1	5
1	1	0	X	6
1	1	1	X	7

4) Characteristic Equation

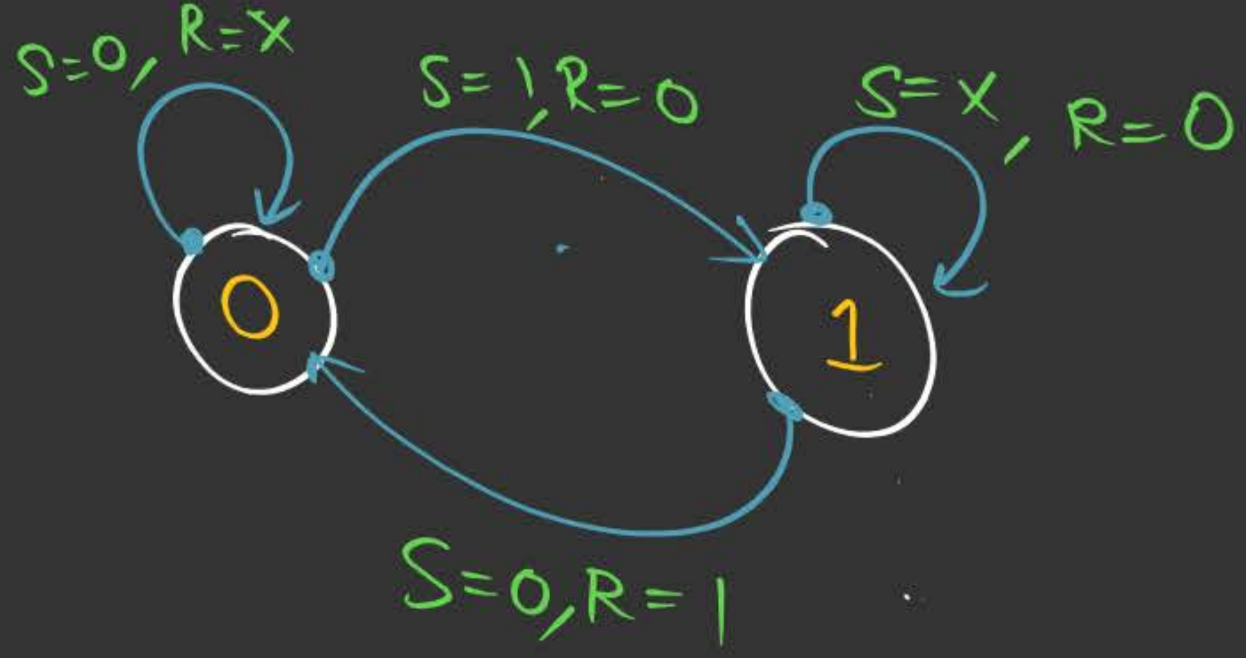
$$\Rightarrow Q_{n+1}(S, R, Q_n) = \Sigma m(1,4,5) + \Sigma d(6,7)$$

$$\boxed{Q_{n+1} = S + \overline{R} \cdot Q_n}$$

5) Excitation table ($Q_n, Q_{n+1} \longrightarrow S, R$)

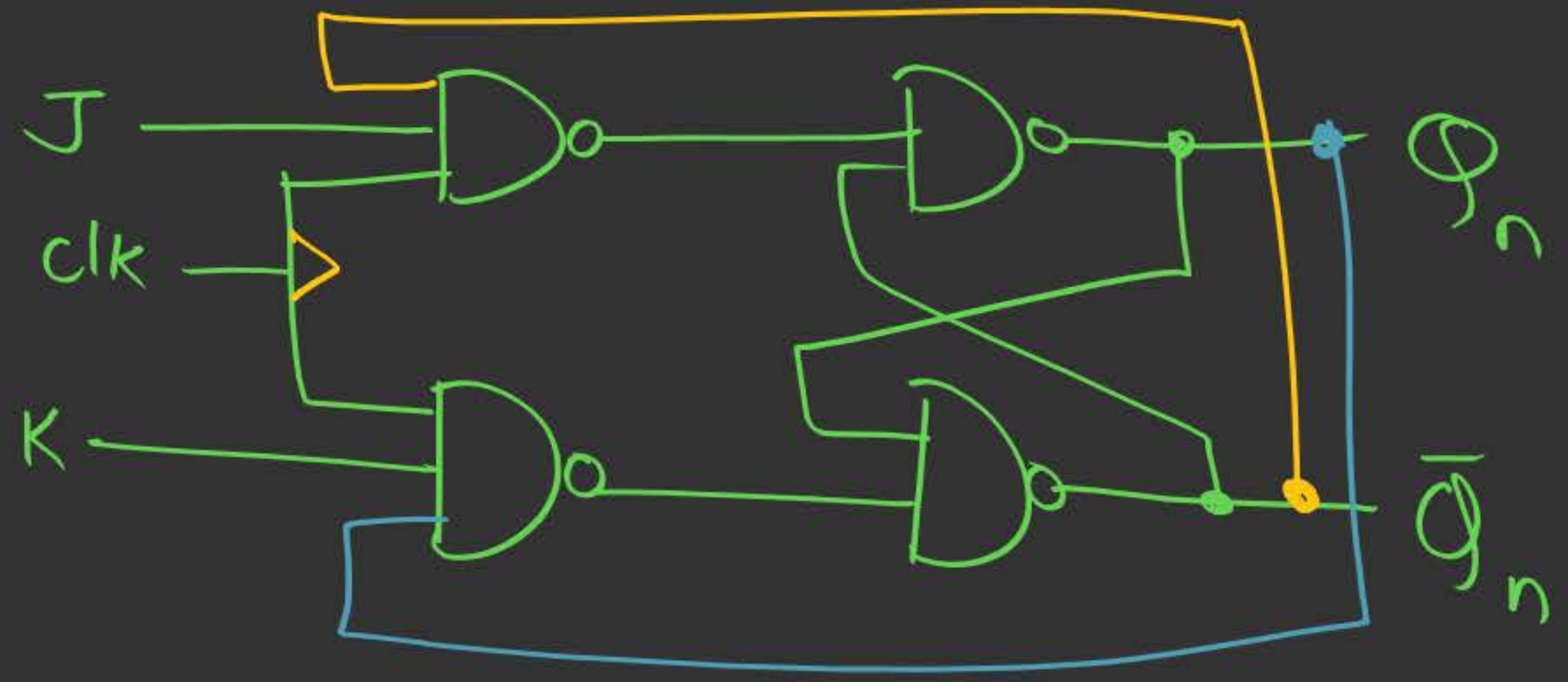
Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

6 State Diagram



② J-K Flip Flop

① Circuit Diagram



② Truth table

J	K	Q_{n+1}	
0	0	Q_n	→ Hold
0	1	0	
1	0	1	
1	1	$\bar{Q}_n$	→ toggle

(2) Truth table

J	K	Q_{n+1}
0	0	Q_n → Hold
0	1	0
1	0	1
1	1	$\overline{Q_n}$ → toggle

(3) Characteristic table ($J, K, Q_n \rightarrow Q_{n+1}$)

J	K	Q_n	Q_{n+1}
0	0	0	→ 0
0	0	1	→ 1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	→ 1
1	1	1	→ 0

(3) Characteristic table ($J, K, Q_n \rightarrow Q_{n+1}$)

J	K	Q_n	Q_{n+1}	
0	0	0	0	0
0	0	1	1	1
0	1	0	0	2
0	1	1	0	3
1	0	0	1	4
1	0	1	1	5
1	1	0	1	6
1	1	1	0	7

(4) Characteristic equation

$$Q_{n+1}(J, K, Q_n) = \sum m(1, 4, 5, 6)$$

KQ_n

J \ KQ_n	$\bar{K}\bar{Q}_n$	$\bar{K}Q_n$	KQ_n	$K\bar{Q}_n$
$\bar{J}$	[0]	1 [1]	[3]	[2]
J	1 [4]	1 [5]	[7]	1 [6]

$J\bar{Q}_n$

$\bar{K}Q_n$

$$\underbrace{J\bar{Q}_n + \bar{K}Q_n}$$

$$\boxed{Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n}$$

characteristic table $(J, K, Q_n \rightarrow Q_{n+1})$

J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

01xx

xx10

⑤ Excitation table

$(Q_n, Q_{n+1} \longrightarrow J, K)$

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Tips to Remember Excitation table (Shortcut)

① S-R FF

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

② JK FF

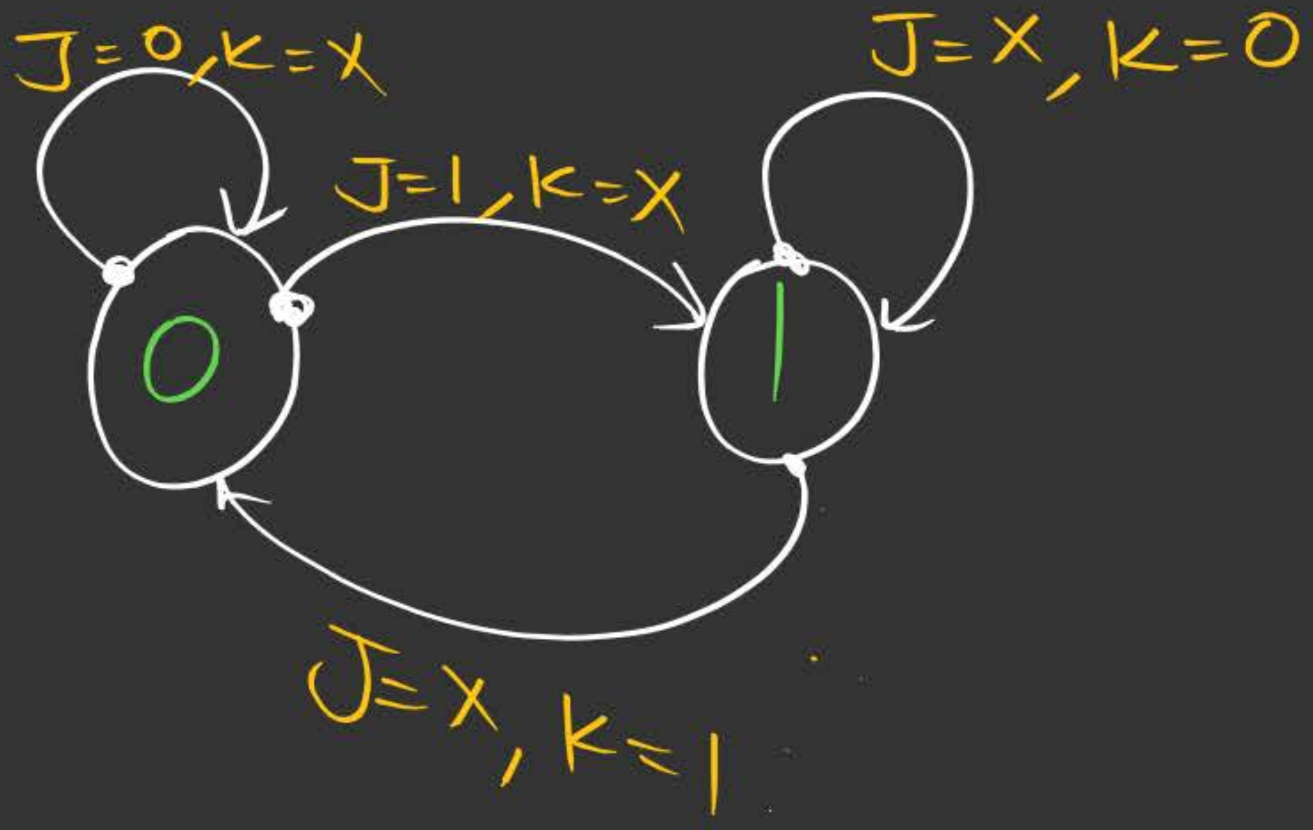
Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

(5) Excitation table

($Q_n, Q_{n+1} \longrightarrow J, K$)

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

(6) State Diagram



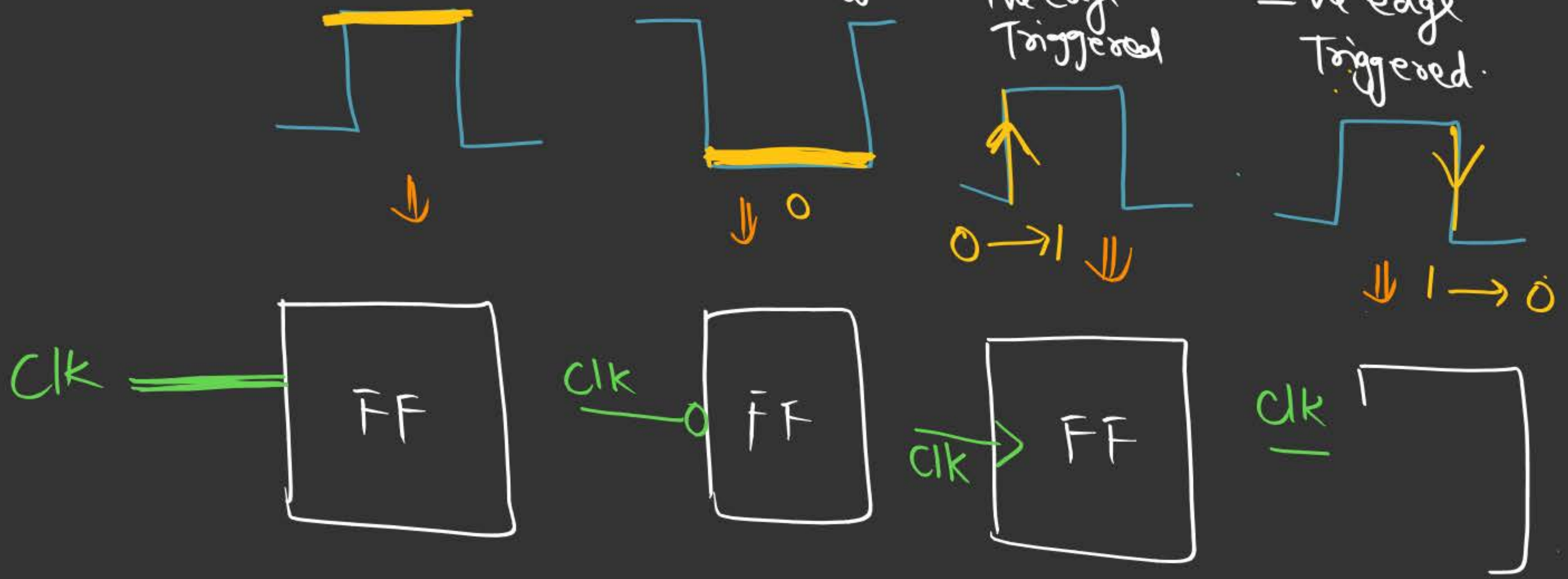
Triggering

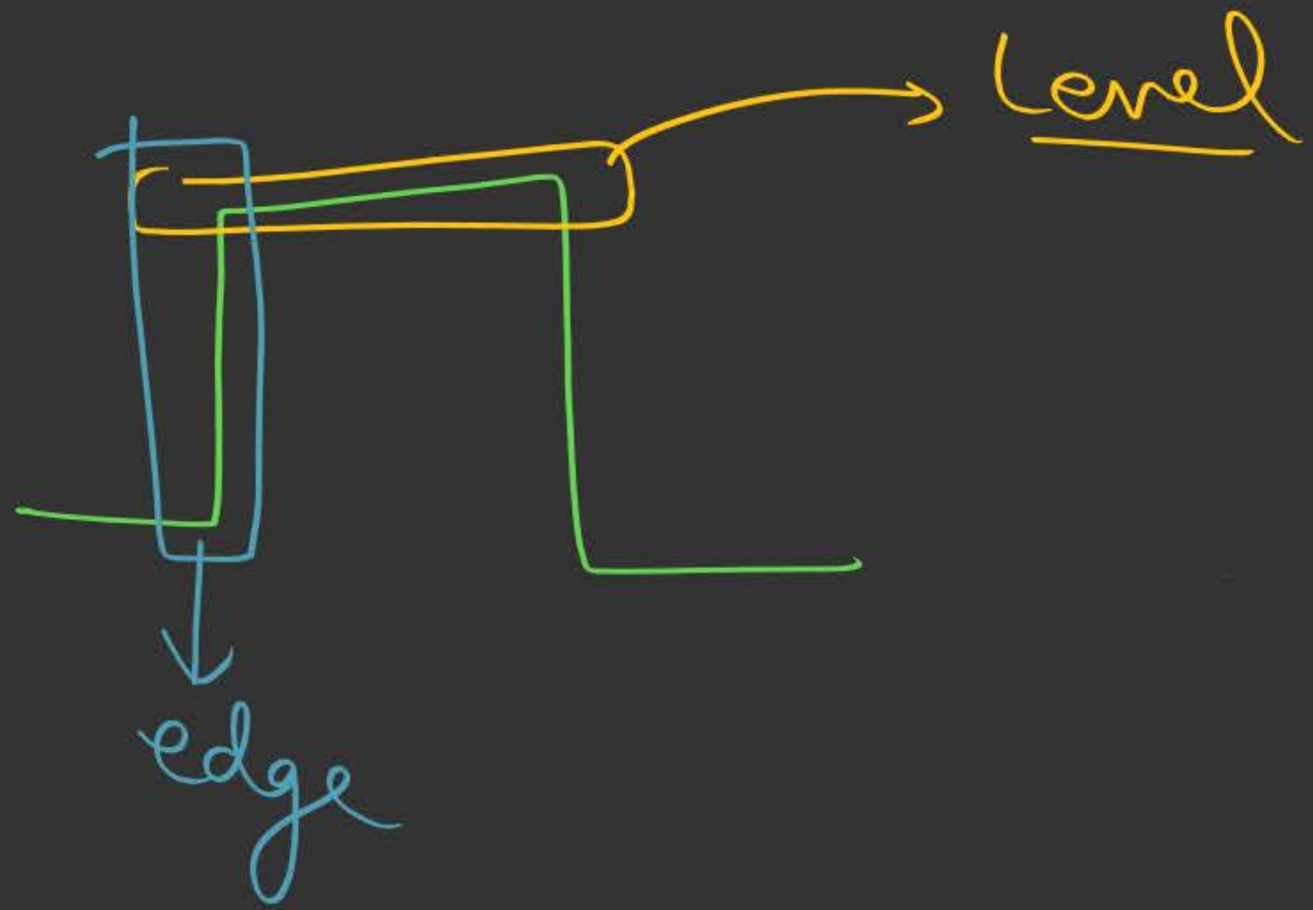
① Level Triggering

- +ve level Triggered
- −ve level triggered

② Edge Triggering

- +ve edge Triggered
- −ve edge Triggered.




Level
edge

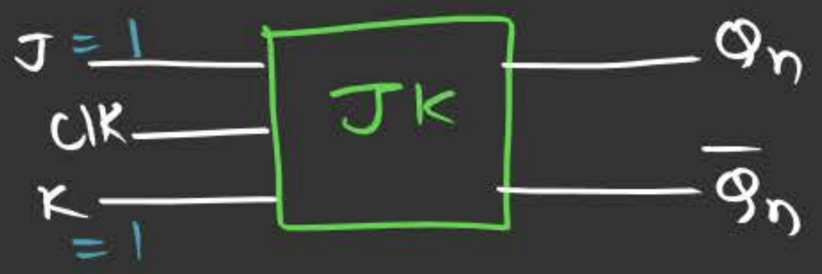
Topic : JK Flip Flop

NOTE: Level triggered JK FF suffers from the problem of RACE AROUND.

Topic : RACE AROUND Problem

When J = K = 1 is applied to the level sensitive of JK FF, then with in the duration of pulse o/p of the FF Toggle more than one times are called RACE AROUND problem.

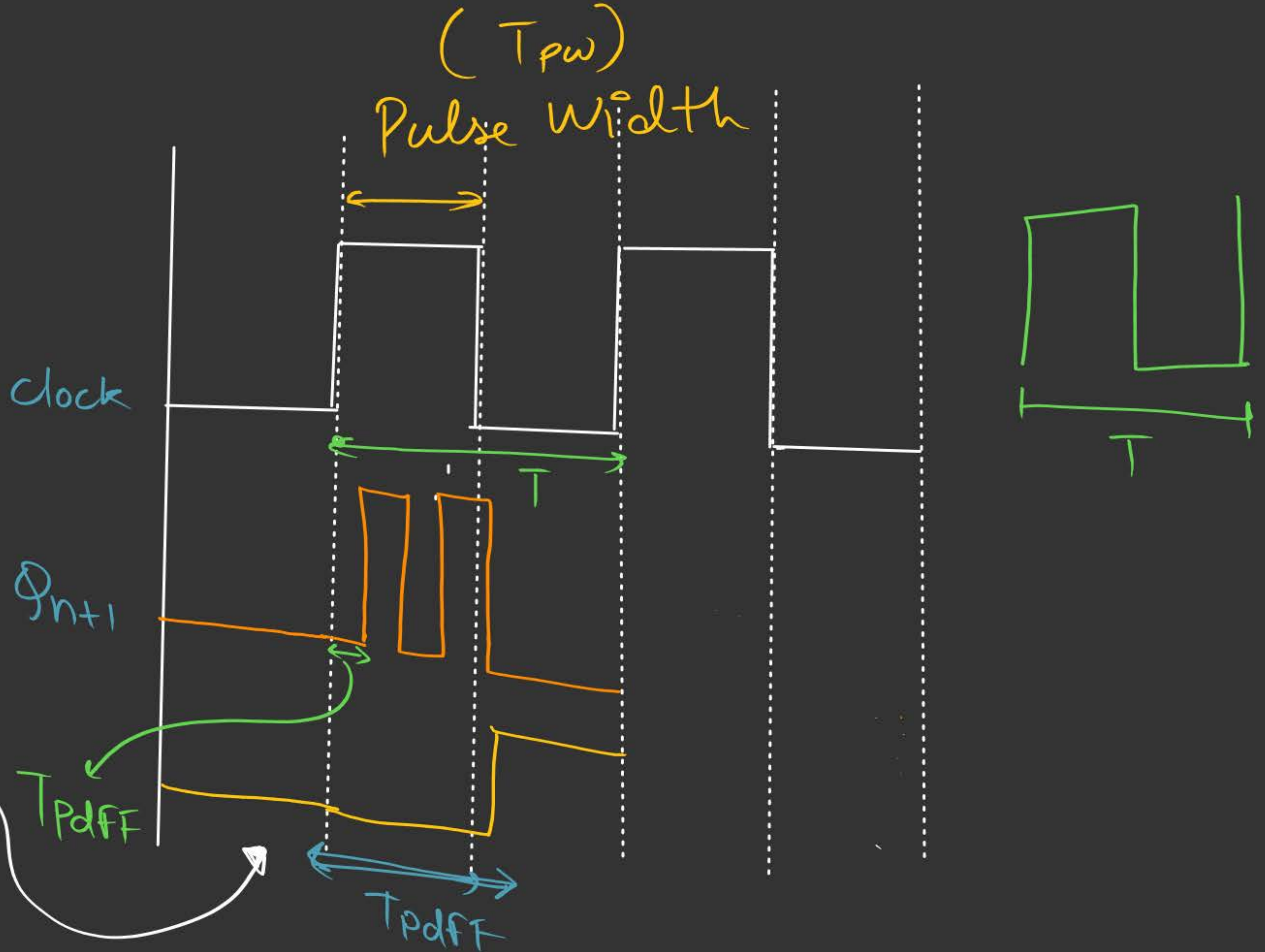
T_{pdFF}



J = 1 and K = 1

Race Around Problem occurs

① $(T_{pw} < T_{pdFF} < T_{clk})$



To avoid the Race Around Problem:-

There are following Solutions:-

1. $(T_{pw} < T_{pdFF} < T_{clk})$
2. Using <u>Master Slave</u> FF.

[MCQ]

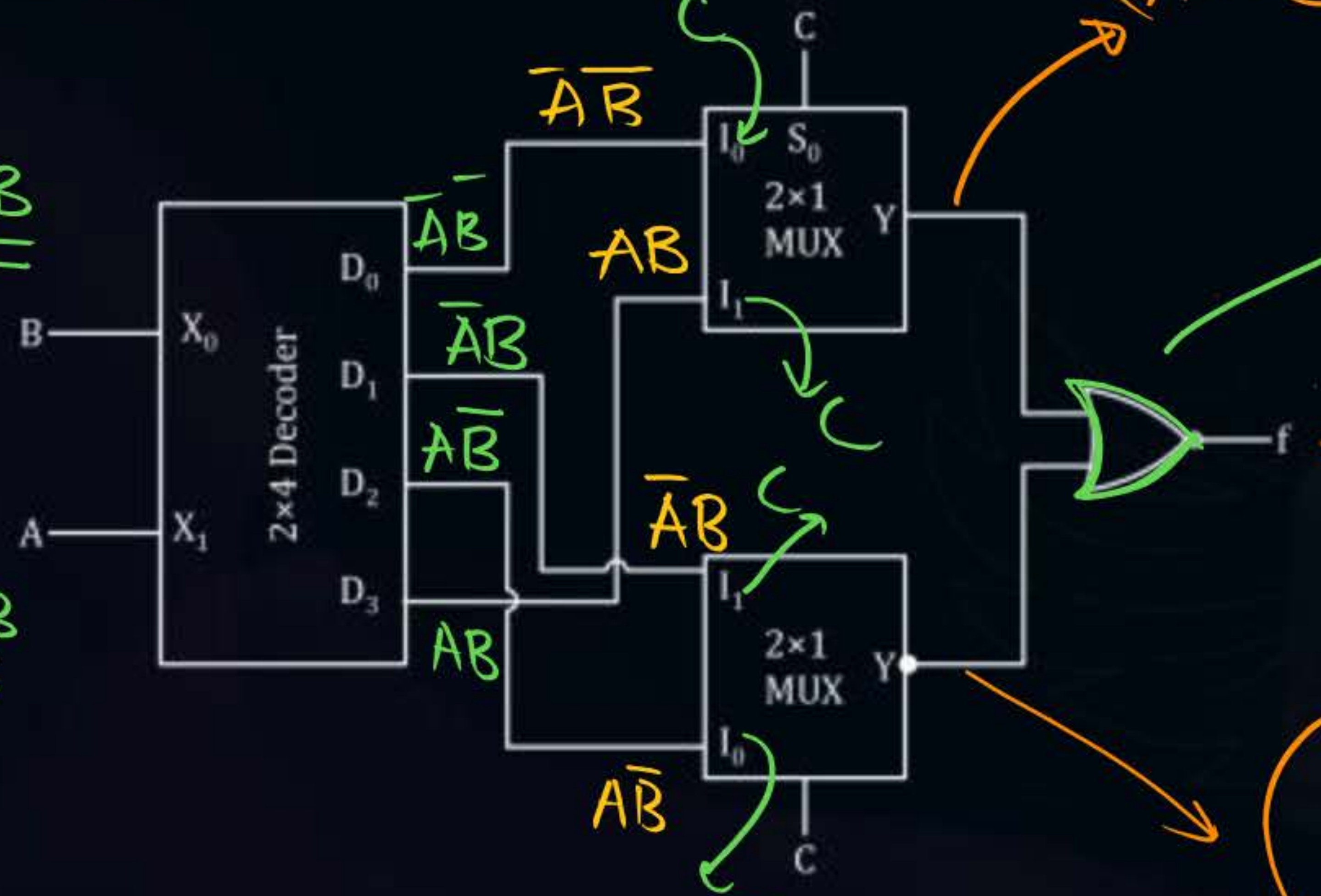
Advanced Question

#Q. A logic function 'f' is implemented by the circuit shown in the figure below. The circuit consists of one 2 × 4 decoder, two 2 × 4 multiplexers and a two input or gate connected in cascade. Then the function f is equal to.

$F(A,B,C)$

LSB

MSB



$(\bar{A}\bar{B}\bar{C} + ABC) \Rightarrow P$

OR

$\Rightarrow P+Q$

$(\bar{A}BC + A\bar{B}\bar{C}) \Rightarrow Q$

A. $A \oplus B$

B. $A \oplus B \oplus C$

C. $B \odot C$

D. $A \odot B$

Appr) :-

$$f(A,B,C) = P + Q$$

$$= (\bar{A}\bar{B}\bar{C} + ABC) + (\bar{A}BC + A\bar{B}\bar{C})$$

$$\begin{array}{cccc} 000 & 111 & 011 & 100 \\ 0 & 7 & 3 & 4 \end{array}$$

$$= \sum m(0,3,4,7)$$

$A \backslash BC$	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	1 $_0$	$_1$	1 $_3$	$_2$
A	1 $_4$	$_5$	1 $_7$	$_6$

$\bar{B}\bar{C}$ BC

$$f(A,B,C) = \bar{B}\bar{C} + BC = B \odot C$$

Appr2:

$$f = P + Q$$

$$= (\bar{A}\bar{B}\bar{C} + ABC) + (\bar{A}BC + A\bar{B}\bar{C})$$

$$= (\bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C}) + (ABC + \bar{A}BC)$$

$$= \bar{B}\bar{C}(\bar{A} + A) + BC(A + \bar{A})$$

$$= \bar{B}\bar{C} + BC$$

$$= B \odot C$$

2 mins Summary

J K FF

Race Around Problem

Imp Question Practice

THANK - YOU